

Geometry

Unit 10

Lesson 6 Homework

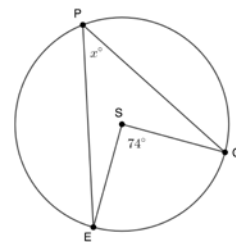
Name Answer Key

Date _____

Period _____

1. Circle S is shown where $\angle GSE$ is a central angle, $\angle GPE$ is inscribed in the circle, and $m\angle GSE = 74^\circ$. Find $m\angle GPE$.

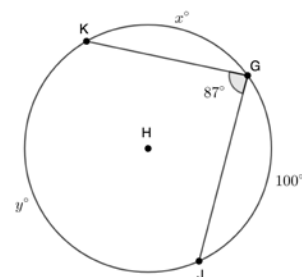
$$\frac{74}{2} = 37^\circ$$



For questions 2-3, Circle H has points K , G , and J on the circle. $\angle K G J$ is inscribed in Circle H measuring 87° , and the $m\widehat{GJ} = 100^\circ$.

2. Find the $m\widehat{KJ}$.

$$87 \cdot 2 = 174^\circ$$



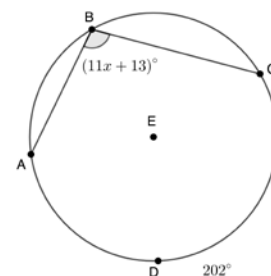
3. Find the $m\widehat{KG}$.

$$360 - 174 - 100 = 86^\circ$$

For questions 4-5, Points A , B , C , and D lie on Circle E . $\angle ABC$ is inscribed in the circle, the $m\angle ABC = (11x + 13)^\circ$, and the $m\widehat{CDA} = 202^\circ$.

4. Find the $m\angle ABC$.

$$\frac{202}{2} = 101^\circ$$



5. Find the value of x .

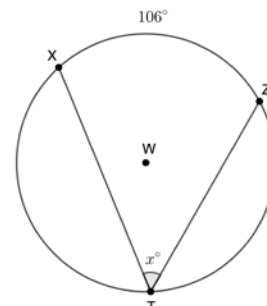
$$11x + 13 = 101$$

$$11x = 88$$

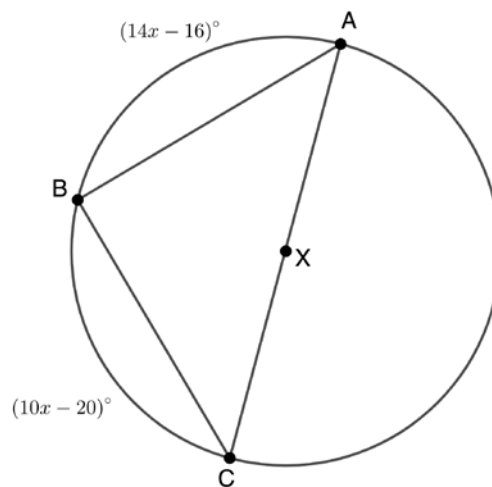
$$x = 8$$

6. Circle W is shown, where points X , Z , and T are on the circle. $\angle XTZ$ is inscribed in Circle W with $m\widehat{XZ} = 106^\circ$. Find the value of x .

$$\frac{106}{2} = 53$$



For questions 7-10, Circle X is shown with inscribed angle $\angle ABC$ and diameter $\overline{CA}$. The $m\widehat{AB} = (14x - 16)^\circ$ and the $m\widehat{BC} = (10x - 20)^\circ$.



7. Find the value of x .
 $14x - 16 + 10x - 20 = 180$
 $24x - 36 = 180$
 $24x = 216$
 $x = 9$

8. Find the $m\angle BAC$.
 $10(9) - 20 = 70$
 $\frac{70}{2} = 35^\circ$

9. Find the $m\angle ACB$.
 $14(9) - 16 = 110$
 $\frac{110}{2} = 55^\circ$

10. Find the $m\widehat{CB}$.
 $10(9) - 20 = 70^\circ$